Third System
AI Presence Index

AI Availability Is Not Recognition Memory

And at n = 24 the familiarity question is still open. A pre-registered discriminant test of AIAS™ Presence against human brand norms, on 24 technology brands.

June 2026 · Pablo Ulpiano González Castro

AIAS™ Measurement Program · Discriminant Validity

AI Availability — whether language models surface, recognize, and recall a brand — only earns its place as a measurable layer of brand growth if it is distinct from the human memory measures it resembles. This study put AIAS™ Presence against two of them, on the hardest substrate for the test to pass: technology, where AI fame and human fame should align most tightly. Presence is distinct from recognition memory. Against familiarity, the evidence at this sample size is genuinely inconclusive — and the pre-registered rule says so rather than pretend otherwise.

rho = 0.38 Presence vs recognition (d') · **rho = 0.59** Presence vs familiarity · **11 / 24** brands dissociate · **n = 24** technology brands

For a new brand metric to be worth the name, it has to measure something the cheaper, older measures do not. This study tested whether AIAS™ Presence — the degree to which a six-model panel recognizes and recalls a brand — is just a re-reading of how well people already know that brand. The validator was the published BRAND database: human familiarity ratings and a signal-detection measure of recognition. The category was chosen to make the test hard. Technology is where model training is densest and where AI fame and human fame should track most closely, so if Presence were going to collapse into a familiarity proxy anywhere, it should collapse here.

Against recognition sensitivity — how reliably people tell a real brand from a decoy — Presence is distinct (rho = 0.38), and its confidence interval rules out the level of correlation that would mark it as redundant. Whether a model classes a brand as "technology" is not a restatement of human recognition. The interval is wide, so this is a confirmed result, not a tight one, and it is reported as such.

Against familiarity — a direct fame rating — the evidence is inconclusive. The central estimate (rho = 0.59) leans toward partial distinctness, exactly as predicted, but across 24 brands the interval is wide enough to span everything from clearly distinct to fully reducible. The pre-registered decision rule returned UNDETERMINED rather than read a finding into a number it cannot support. The fame question is open, not answered — and a larger panel is what would close it.

Where Presence and familiarity diverge, they diverge in a pattern. Eleven of 24 brands pull apart by more than a standard deviation, in both directions: the model layer amplifies core-technology identity — Apple, Microsoft, and the enterprise infrastructure names — and suppresses brands that are culturally famous without being category-canonical technology, such as Netflix, Instagram, Uber, and Airbnb. AI presence and human fame are not the same map of a category. The difference is legible, not noise, and it is the part of this result with the most immediate use for anyone managing a brand's standing inside the models.

What we measured

Presence was composed from a fixed six-model panel in two phases. The recognition phase asked each model, for each brand, whether it is commonly recognized as a technology brand — a yes/no count from zero to six. The recall phase posed six category-level prompts and counted how often each brand surfaced, split into a quality channel (best, expert-chosen, highest-quality) and a cultural channel (most talked-about, biggest footprint, most iconic). Presence is the equal-weight mean of those three components on a 0–100 scale — the same composition behind the program's convergent benchmark.

The 24-brand panel was drawn from the technology category of the BRAND database, stratified across the full familiarity range so the test would not be confined to famous brands or obscure ones. It spans Apple, Google, and Microsoft at one end through Corning, VMware, NetApp, Jabil, and Pitney Bowes at the other. One Samsung-owned brand was removed before sampling to keep the panel clean of any conflict.

The whole measurement was run blind to the human validator. Presence was composed first; only at a single, pre-registered scoring step was it joined to familiarity and recognition and the verdicts read. Nothing about the human norms could shape how Presence was built.

FINDING 01

The recognition question is settled; the fame question is open

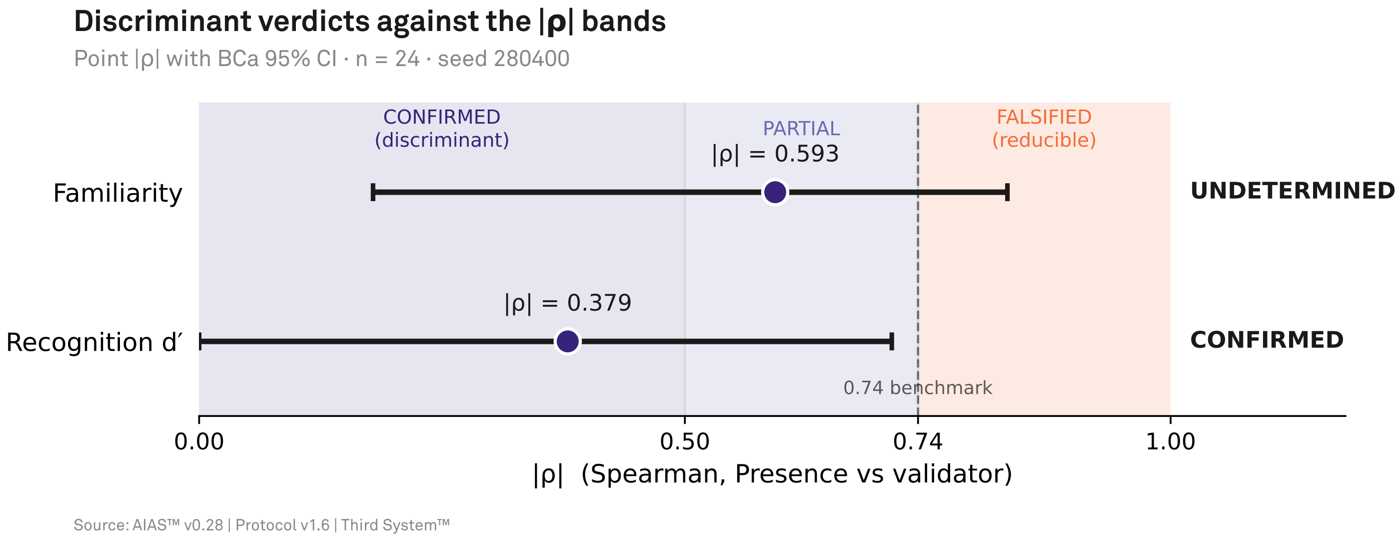


Figure 1 · Discriminant verdicts against the convergent benchmark. |rho| for AIAS Presence vs. each BRAND human-norm measure, with BCa 95% confidence intervals, against the three decision bands (CONFIRMED |rho| < 0.50; PARTIAL 0.50–0.74; FALSIFIED >= 0.74, the v0.25 convergent anchor). Recognition memory (d'): rho = 0.38, CONFIRMED. Familiarity: rho = 0.59, UNDETERMINED — the CI [0.18, 0.83] spans all three bands at n = 24.

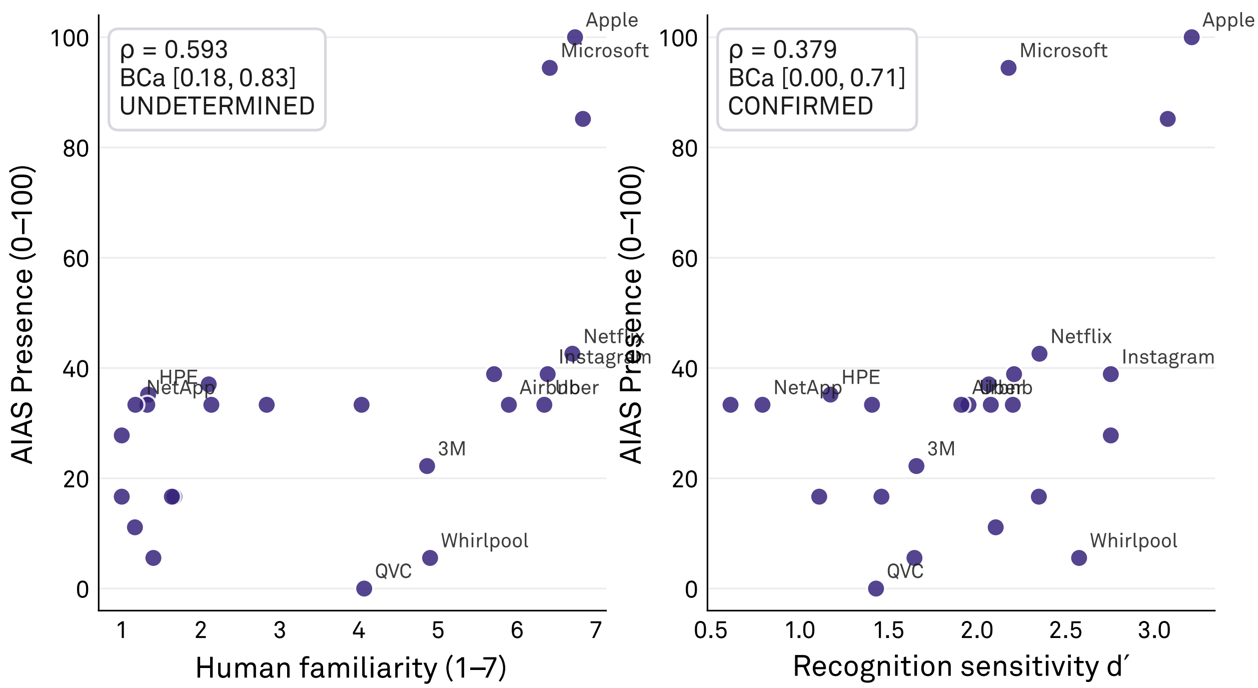
Two relationships were tested against a fixed reducibility threshold of 0.74 — the level at which a discriminant correlation would be as strong as the metric's own convergent benchmark, and therefore redundant. Presence versus recognition sits well inside the distinct range and its interval stops short of that line: confirmed. Presence versus familiarity has its central estimate in the partial range, but an interval so wide it reaches from clearly-distinct to fully-reducible. The locked rule reads that as unresolved, not as a partial finding. The figure shows why: one interval clears the threshold, the other straddles the whole field.

FINDING 02

Presence and human memory track each other loosely

Presence vs human brand norms

Discriminant validity · two gating relationships · n = 24



ρ is rank-based (Spearman); axes show raw values. Labelled points are the dissociation crossers.

Source: AIAS™ v0.28 | Protocol v1.6 | Third System™

Figure 2 · Presence vs. human brand norms. AIAS Presence (0–100) against BRAND familiarity (1–7, left) and recognition sensitivity d' (right) across the 24 Technology-panel brands. Both relationships sit below the convergent ceiling: Presence is distinct from recognition memory (CONFIRMED) and inconclusively related to familiarity at this sample size (UNDETERMINED).

Plotted against both human norms, Presence rises with them but scatters widely around the trend. The leaders cluster high on every measure; the rest of the panel spreads out, and the brands that sit far off the line are the same ones that dissociate. The loose fit is the point — a tight fit would have meant Presence was simply re-indexing fame.

FINDING 03

The model layer keeps its own map of the category

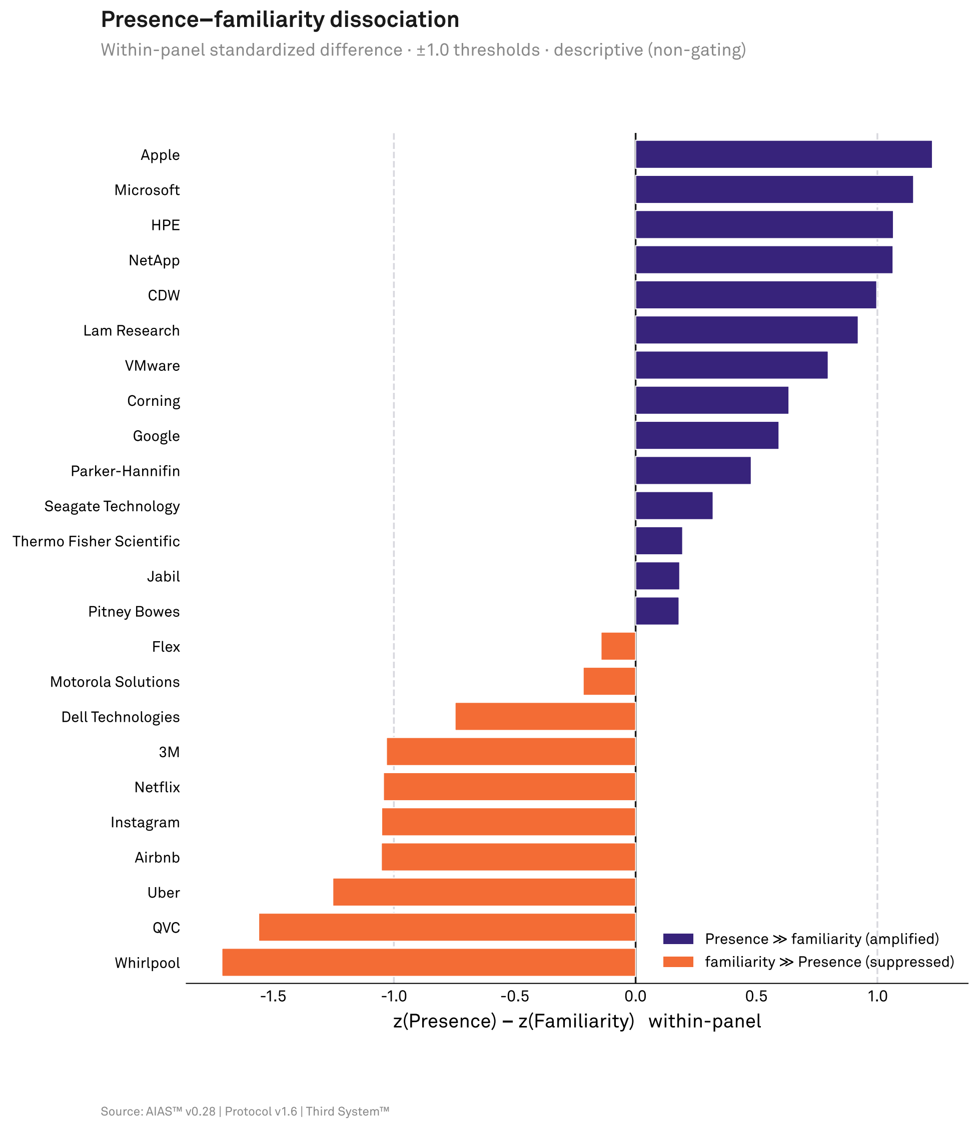


Figure 3 · Where Presence and familiarity part ways. Within-panel z(Presence) - z(familiarity) for all 24 brands. Eleven brands cross the ±1.0 threshold in both directions (FULL dissociation): Presence-amplified (Apple, Microsoft, HPE, Netapp) and familiarity-amplified (Netflix, Instagram, Uber, Airbnb, Whirlpool, 3M, QVC). Descriptive; illustrates the primary correlation rather than testing it independently.

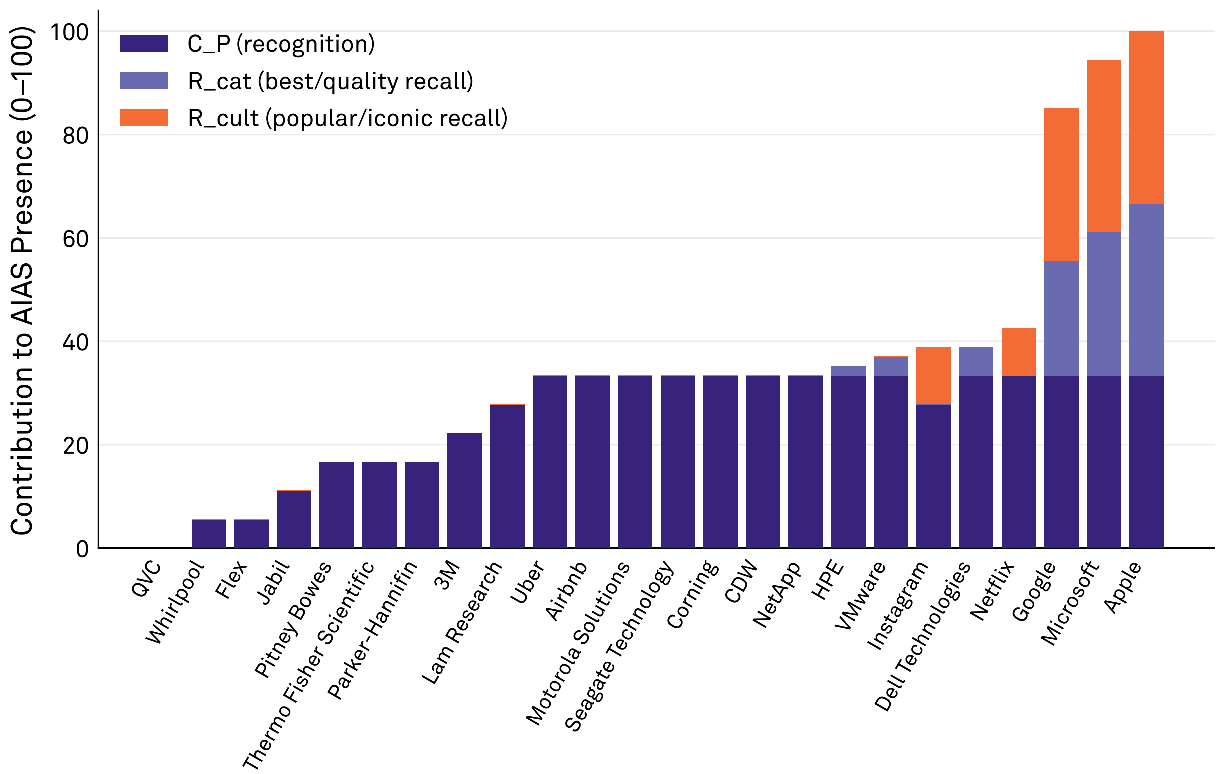
Eleven brands diverge by more than a standard deviation between AI presence and human familiarity. The amplified set — Apple, Microsoft, HPE, NetApp — reads larger in the models than in human fame: core-technology identity the panel surfaces readily. The suppressed set splits in two: culturally famous brands the models do not treat as category-canonical technology (Netflix, Instagram, Uber, Airbnb), and human-familiar brands with thin model presence (Whirlpool, 3M, QVC). Knowing which side a brand falls on is the actionable read here.

FINDING 04

For enterprise brands, AI recall goes quiet

Presence composition by component

Recall floors for the enterprise-heavy panel; Presence reduces toward C_P (recognition)



Source: AIAS™ v0.28 | Protocol v1.6 | Third System™

Figure 4 · What drives each brand’s Presence. Per-brand Presence decomposed into its three v1.6 components — Recognition C_P (/6), category-recall R_cat (/18), and cultural-recall R_cult (/18), each scaled to 0–100 and averaged. Enterprise brands rest almost entirely on Recognition with little recall; only a few consumer-facing brands (Apple, Microsoft, Google) carry both recall channels.

Presence has three parts: recognition, quality-recall, and cultural-recall. For most of this enterprise-heavy panel the two recall channels contribute almost nothing — the models recognize the brand as technology but do not volunteer it under 'best' or 'most iconic' consumer prompts. Presence reduces toward its recognition component, which is the main reason the familiarity test came in underpowered: for these brands, the recall signal that would most sharply distinguish Presence from fame simply did not fire.

Limitations

The binding limitation is sample size. At 24 brands the confidence intervals are wide enough that the primary, familiarity question cannot be resolved either way. The panel size was fixed by the need to stratify a single category by familiarity while holding the model panel and prompt battery constant; resolving the fame question requires a larger panel, which is the next study, not a reinterpretation of this one.

The substrate amplified that constraint. An enterprise-heavy technology panel under consumer-framed recall prompts floored the recall channels, so for most brands Presence reduced toward recognition. The discriminant tests therefore largely evaluated AI recognition against the human norms, not the full three-part Presence construct. A consumer category, where recall does not floor, would test the whole construct against familiarity more cleanly.

The reducibility threshold is borrowed from the program's convergent benchmark, which used a wider prompt set than the battery here, so the comparison is approximate on prompt breadth. It is used as a fixed, pre-registered decision line, not as a precise like-for-like quantity.

What’s next

The immediate next step is a higher-powered replication of the familiarity test on a consumer category where recall does not floor, so that the full Presence composite — not its recognition component alone — is what confronts fame. That is what would convert this open question into a verdict.

Two further tests would complete the picture: a direct comparison against mental availability using category-entry-point measures, and a recognition replication on a substrate where AI and human recognition are expected to diverge more sharply. Consolidating these is the gate to expanding AIAS beyond Presence to the full multi-component score.

This study is one cell of a larger validation. It sits alongside the convergent evidence (Presence tracks an external interest signal) and the discriminant evidence against retail-sales rank, together forming a convergent-and-discriminant pairing across methods. Read as a set, the program is building the case that AI Availability measures something the existing brand measures do not.

Hypothesis scoring

Three pre-registered hypotheses, scored once against the locked verdicts on the frozen 24-brand Technology panel (seed 280400; blinding released only at scoring).

H	Pre-registered prediction	Result	Status
H_Disc_Familiarity	Presence vs familiarity (primary)	rho = 0.593, central estimate in the partial band; BCa 95% CI [0.179, 0.832] spans all three bands, so the locked override returns UNDETERMINED rather than the point verdict. Unresolved at n = 24.	UNDETERMINED
H_Disc_Recognition	Presence vs recognition d' (secondary)	rho = 0.379, inside the distinct band; BCa 95% CI [-0.076, 0.714], upper bound below the 0.74 reducibility threshold, so redundancy is excluded. Confirmed, though the interval is wide.	**CONFIRMED**
H_Dissociation	Presence–familiarity dissociation (descriptive)	11 of 24 brands cross ±1 SD in both directions (4 amplified, 7 suppressed). Mechanically coupled to the primary correlation; reported as corroborating texture, excluded from the headline.	FULL

Detailed hypothesis results

H_Disc_Familiarity — UNDETERMINED. The point estimate landed where the registered prediction said it would, in the partial band. What the prediction could not anticipate was how wide the interval would be at this sample size: from 0.18, which would mean a clearly distinct measure, to 0.83, which would mean one effectively reducible to fame. An honest reading cannot pick a side, and the pre-registered rule was written precisely to stop a tempting central number from being reported as a result. The verdict is "unresolved," and it is the most defensible thing the data support.

H_Disc_Recognition — CONFIRMED. Presence correlates with human recognition sensitivity only moderately, and the interval's upper bound stops short of the redundancy threshold. The model's judgement that a brand is "technology" is not a restatement of how reliably a person recognizes it. The interval reaches into the partial band, so the discriminance is confirmed but not robust — stated plainly rather than rounded up.

H_Dissociation — FULL. The amplified brands are the category's core-technology names; the suppressed brands are either culturally famous but not category-canonical, or human-familiar with thin model presence. The pattern is systematic and reconciles with the recall evidence — the brands the models recall culturally but not for quality, and the reverse. It is descriptive and mechanically tied to the familiarity correlation, so it corroborates rather than proves, but it is the clearest picture of how AI presence and human fame part ways.

About the Third System

The Third System describes a shift in how consumers make decisions: instead of browsing and comparing options directly, many now rely on AI systems to generate recommendations. In this environment, brands compete not only for human attention but for inclusion within AI-generated outputs. This creates a new competitive layer where visibility is determined by how AI systems interpret and surface brands.

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Methodology

This report presents findings from Third System's AI Availability research program. The AI Availability Score (AIAS) is an early-stage measurement framework introduced in Tri-System Brand Growth (Gonzalez Castro, 2026). Findings should be used directionally. Construct validity — the correlation between AI Availability and consumer behavior — has not yet been established and is the subject of Phase 3 of the research program. Methodology is published in full and versioned at thirdsystem.ai/methodology (current: v1.6). Registry construction is curated and reflects the framework's view of each category's competitive set; revisions are documented in the methodology log.

Citation

González Castro, P. U. (2026). ***AI Availability Is Not Recognition Memory — and Is Underpowered Against Familiarity at n = 24: A Discriminant-Validity Test of AIAS Presence Against Human Brand Norms (v0.28 / CV.04)***. Third System. thirdsystem.ai/v28

Underlying datasets

Human norms: the BRAND database (familiarity and recognition d', Brand Finance US 500). AI Presence: a six-model panel, 144 recognition probes and 36 recall queries over 24 technology brands. Pre-registration, data, scoring, and verdicts deposited at OSF (osf.io/ec6wh, v28).

Methodology log: Pre-registered at git tags v0.28-prereg-r1 / r2 before any model was queried. Presence composed blind to the validator; joined to the human norms only at a single, one-shot scoring step (seed 280400, BCa 95% CI, 10,000 resamples). Verdicts locked at v0.28-results-locked and deposited unchanged..

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